Code. RESOLUTION MEPC.259(68) (adopted on 15 May 2015) 2015 GUIDELINES FOR EXHAUST GAS CLEANING SYSTEMSMEPC 68/21/Add.1 Annex 1, page 20 Documents/English/MEPC 68-21-ADD.1 (E).doc Table 1: Fuel properties for marine distillate and residual fuel* Carbon Hydrogen Sulphur Other C H S Fuel S/C Exh SO2/CO2 Fuel Type %(m/m) %(m/m) %(m/m) %(m/m) mol/kg mol/kg mol/kg mol/mol ppm/%(v/v) Distillate 86.20 13.60 0.17 0.03 71.8333 136 0.0531 0.00074 7.39559 Residual 86.10 10.90 2.70 0.30 71.7500 109 0.8438 0.01176 117.5958 Distillate 85.05 13.42 1.50 0.03 70.8750 134.2 0.4688 0.006614 66.1376 Residual 87.17 11.03 1.50 0.30 72.6417 110.3 0.4688 0.006453 64.5291 * Based on properties in the IMO NOx Monitoring Guidelines, resolution MEPC.103(49). Table 2: Emissions calculations corresponding to 1.5 % fuel sulphur CO2 SO2 Exh SO2/CO2 Exh S/C % ppm4 ppm4/% m/m Distillate 0.17% S 8 59.1 7.4 0.00197 Residual 2.70% S 8 939.7 117.5 0.03136 Distillate 1.5% S 8 528.5 66.1 0.01764 Residual 1.5% S 8 515.7 64.5 0.01721 Distillate 1.5% S 0.5 33.0 66.1 0.01764 Residual 1.5% S 0.5 32.2 64.5 0.01721 - Residual 2.70% S 8 939.7 117.5 0.03136 Distillate 1.5% S 8 528.5 66.1 0.01764 Residual 1.5% S 8 515.7 64.5 0.01721 Distillate 1.5% S 0.5 33.0 66.1 0.01764 Residual 1.5% S 0.5 32.2 64.5 0.01721 SO2/CO2 ratio vs % sulphur in fuel 0 50 100 150 200 250 0,00 0,50 1,00 1,50 2,00 2,50 3,00 3,50 4,00 4,50 5,00 % m/msulphur in fuel SO2(ppm)/CO2(%) Distillate fuel Residual fuel RESOLUTION MEPC.259(68) (adopted on 15 May 2015) 2015 GUIDELINES FOR EXHAUST GAS CLEANING SYSTEMSMEPC 68/21/Add.1 Annex 1, page 21 Documents/English/MEPC 68-21-ADD.1 (E).doc 4 Correspondence between 65 (ppm 4/%) SO2/CO2 and 6.0 g/kWh is demonstrated by showing that their S/Cratios are similar. This requires the additional assumption of abrake-specified fuel consumption value of 200 g/kWh. This is an appropriate average for marine diesel engines. The calculation is as follows: S/Cfuel = brake-specific SO2 (MWS / MWSO2) BSFC (% carbon in fuel / 100) brake-specific SO2 = 6.0 g/kW-hr MWS = 32.065 g/mol MWSO2 = 64.064 g/mol BSFC = 200 g/kW- hr % carbon in 1.5% sulphur fuel (from table 1) = 85.05% (distillate) or 87.17% - (MWS / MWSO2) BSFC (% carbon in fuel / 100) brake-specific SO2 = 6.0 g/kW-hr MWS = 32.065 g/mol MWSO2 = 64.064 g/mol BSFC = 200 g/kW- hr % carbon in 1.5% sulphur fuel (from table 1) = 85.05% (distillate) or 87.17% (residual) S/Cresidual fuel = 6.0 (32.065 / 64.064) = 0.01723 200 (87.17% / 100) S/Cdistillate fuel = 6.0 (32.065 / 64.064) = 0.01765 200 (85.05% / 100) Note 1: The S/Cmass ratios calculated above, based on 6.0 g/kWh and 200 g/kWh BSFC, are both within 0.10% of the S/Cmass ratios in the emissions table (Table 2). Therefore, 65 (ppm4/%) SO2/CO2 corresponds well to 6.0 g/kWh. Note 2: The value of 6.0 g/kWh, hence the 200g/kWh brake-specified fuel consumption is taken from MARPOL Annex VI as adopted by the 1997 MARPOL Conference. 5 Thus, the working formulas are as follows: For complete combustion = SO2 (ppm*) ■ 65 CO2 (%*) For complete combustion = SO2 (ppm*) ■ 65 CO2 (%*) +